Some text before a display of a formula, which is set in display style with the equation number at the right:

$$\int_0^\infty e^{-x^2}\,dx = \frac{\sqrt{\pi}}{2}. \tag{1}$$

Then the text goes on, and another display follows with a sum and a fraction, its number at the left:

(2) $\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$

and a third without a number, with delimiters and a radical:

$$\begin{bmatrix} a \\ c \end{bmatrix} \qquad \sqrt{x^2 + y^2} \leq |x| + |y|.$$

More text, enough to go on to a second page with more displays. More text, enough to go on to a second page with more displays. More text, enough to go on to a second page with more displays.

$$f(x) = \overline{x^2} + \underline{\hat{x}} \cdot \tilde{y}\left(z_1^2\right)$$

$$\lim_{n \to \infty} \left(1 + \frac{1}{n}\right)^n = e$$

The end of the text with displays, after which the document ends.

$$\prod_{p \text{ prime}} \frac{1}{1 - p^{-s}} = \sum_{n \geq 1} n^{-s} \tag{3}$$